

CUSTOMER MILESTONE BRIEF

MS #08 - Runtime Evidence Plane

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.8.0
Date	2026-08-11
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #8 - Runtime Evidence Plane converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Adds observed runtime evidence to the static estate map and distinguishes observed, contradicted, and static-only claims.

Why this matters to a customer

Adds observed runtime evidence to the static estate map and distinguishes observed, contradicted, and static-only claims.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a versioned runtime-evidence contract that binds every observation to an existing graph node or edge and rejects unknown identifiers before ingestion.
- Added replaceable local-oracle and recorded-fixture adapters plus a clean boundary for future z/OSMF, JES, spool, SMF, and dataset-allocation collectors.
- Added append-only SHA-256 event chains, content-addressed run receipts, deterministic compressed snapshots, source-system identity, artifact hashes, and declared limitations.
- Added runtime reconciliation with static_only, runtime_observed, and runtime_contradicted states and evidence-class-aware confidence scoring.
- Added separate development-readiness and mainframe-equivalence policies; simulated and local evidence can exercise the factory but only zos_observed evidence can prove mainframe parity.

- Added a Runtime explorer view for adapter runs, trust policies, observed operations, limitations, receipt identities, and node/edge runtime projections.
- Added deterministic build and verification launchers, JSON Schemas, a synthetic z/OS-shaped replay fixture, tamper and contradiction tests, and full-suite integration.

How it differs from earlier work

MS #8 builds on MS #7 (Autonomous Factory Run Engine). The earlier milestone established its own bounded capability; this milestone adds runtime evidence plane while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use runtime evidence plane as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.8.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.